Type MS · Describe how heart disease is prevented and treated

Answer 1 of 7 · Question 4 part a · 2 marks

2023 NOV · Paper 02 · extract 1 of 2 · find part a in this mark scheme extract

4 (a)	• reduce fat/cholesterol intake; • increase exercise;		2
(b) (i)	Any four from • liver breaks down toxins/drugs; • (if not working properly) statins won't be broken down/will accumulate/reduced detoxification; • kidneys excrete waste (products)/named waste products; • (if not working) can't be removed/less waste/named waste filtered/removed; • increased risk of (further) liver/kidney damage;	Ignore references to deamination	4
(ii)	• statins could cross from mother to fetus; • through the placenta; • damage/affect growth/development of fetus/lead to low birth weight of fetus; • could be in mother's milk if breastfeeding;	Allow baby for mp's 1 and 3 Ignore affects fetus	3
(c) (i)	• (16.3 : 8) = 2.0;	Accept 2.0375 Accept 2/2.04	1
(ii)	Any four from • Double the number/more symptoms shown with patients taking statins than those taking no tablets • even on statins symptoms not high; • almost same with placebo/symptoms with placebo similar/just below those on statins; • symptoms still present when taking no statins; • statins not the cause/other causes of symptoms/named cause e.g. age/gender/fitness/lifestyle;		4 exp

Total 14 marks

Type MS · Describe how heart disease is prevented and treated

Answer 1 of 7 · Question 4 part a · 2 marks

2023 NOV · Paper 02 · extract 2 of 2 · find part a in this mark scheme extract

Question number	Answer	Notes	Marks
			2

Type MS · Describe how heart disease is prevented and treated

Answer 2 of 7 · Question 6 part c.iii · 1 mark

2020 JAN · Paper 01 · find part c.iii in this mark scheme extract

Number				
6	(a)	bronchus/bronchi (1)		1
	(b) (i)	• inhale and exhale/breathe in an out (1) • through mouthpiece (1) • floating air chamber rises/falls (1) • pen records a trace on (the rotating) graph paper/drum (1)		4
	(ii)	Do not share the mouthpiece with another person/disinfect mouthpiece before use by another person/after use (1)		1
	(iii)	Any two from: • airways /bronchi/bronchus constricted/obstructed/reduced diameter (1) • less/reduced airflow (1) • per unit time (1)		2
	(c) (i)	cannot be transmitted from one person to another/not contagious/not caused by a pathogen/bacteria/virus/microorganism (1)		1
	(ii)	9.5 - 7 = 2.5(1) 2.5 ÷ 7 x 100(1) 36(%) (1)	Full marks for correct final answer Allow 35.7 for final answer	3
	(iii)	people taking less exercise/ consuming too much fat/sugar/ taking in more energy than is being used/ increase in the number of people smoking/drinking more alcohol/obesity (1)	Allow other valid lifestyle factor	1

Total for Question 6 = 13 marks

Type MS · Describe how heart disease is prevented and treated

Answer 3 of 7 · Question 4 part c.i · 2 marks

2024 NOV · Paper 02 · extract 1 of 2 · find part c.i in this mark scheme extract

4 (a)	• diarrhoea by clean water/safe water/ bottled water/boiled water/ sewage treatment/chlorination/good sanitation/not mixing sewage with drinking water/washing hands/good hygiene/ cooking food thoroughly; • malaria by nets/draining swamps/insecticides/ insect repellent's/wearing clothing to cover arms (and legs/mosquito coils/anti-malarial/vaccination ;	Ignore ORT	2
(b) (i)	antibiotics;		1
(ii)	• elderly have weakened/lowered immune system/immunocompromised; • indoor spaces makes transmission easier/reduced air flow/ reduced ventilation/build- up of air droplets with pathogens/ people closer together;		2
(c) (i)	• diet; • exercise; • smoking: • drinking alcohol: • stress	Accept drugs	2
(ii)	• $\frac{5\,000\,000}{100\,000} = 50$; • $50 \times 445 = 22\,250$; • 22 300; ***22,250 scores 2 marks*** ***22,300 scores 3 marks***	***1 mark for ×445 if there are no other marks***	3
(iii)	• use of 445 and 185; • $\frac{445}{185}$; • 2.41:1/2.405:!/2.4:1= 3 marks;	ecf from mp1 **if they have done the correct calculation and 2:1 is on the answer line they score** 2 marks	3
(iv)	• people live longer/elderly population; • these diseases are more common in older people; • less likely to die younger from infectious diseases/better medical facilities/treatment; • different diet/ more meat/saturated fats/processed food; • drinking more alcohol;	People live longer so there is more cancer and dementia/these diseases= 2 marks	2

Total 15 marks

Type MS · Describe how heart disease is prevented and treated

Answer 3 of 7 · Question 4 part c.i · 2 marks

2024 NOV · Paper 02 · extract 2 of 2 · find part c.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Describe how heart disease is prevented and treated

Answer 4 of 7 · Question 4 part c.ii · 4 marks

2023 NOV · Paper 02 · extract 1 of 2 · find part c.ii in this mark scheme extract

4 (a)	• reduce fat/cholesterol intake; • increase exercise;		2
(b) (i)	Any four from • liver breaks down toxins/drugs; • (if not working properly) statins won't be broken down/will accumulate/reduced detoxification; • kidneys excrete waste (products)/named waste products; • (if not working) can't be removed/less waste/named waste filtered/removed; • increased risk of (further) liver/kidney damage;	Ignore references to deamination	4
(ii)	• statins could cross from mother to fetus; • through the placenta; • damage/affect growth/development of fetus/lead to low birth weight of fetus; • could be in mother's milk if breastfeeding;	Allow baby for mp's 1 and 3 Ignore affects fetus	3
(c) (i)	• (16.3 : 8) = 2.0;	Accept 2.0375 Accept 2/2.04	1
(ii)	Any four from • Double the number/more symptoms shown with patients taking statins than those taking no tablets • even on statins symptoms not high; • almost same with placebo/symptoms with placebo similar/just below those on statins; • symptoms still present when taking no statins; • statins not the cause/other causes of symptoms/named cause e.g. age/gender/fitness/lifestyle;		4 exp

Total 14 marks

Type MS · Describe how heart disease is prevented and treated

Answer 4 of 7 · Question 4 part c.ii · 4 marks

2023 NOV · Paper 02 · extract 2 of 2 · find part c.ii in this mark scheme extract

Question number	Answer	Notes	Marks
			2

Type MS · Describe how heart disease is prevented and treated

Answer 5 of 7 · Question 3 part b.iii · 2 marks

2020 NOV · Paper 02 (regional R) · extract 1 of 2 · find part b.iii in this mark scheme extract

Number				
3 (a)	(i)	A (1) (aorta) B it is not the pulmonary artery C it is not the pulmonary vein D it is not the vena cava		1
	(ii)	C (1) (right atrium) A it is not connected to the left atrium B it is not connected to the left ventricle D it is not connected to the right ventricle		1
(b)	(i)	circle drawn beyond by-pass vessel and around end of coronary vessels (1)		1
	(ii)	• no/less blood to heart muscles (1) • no/less oxygen/glucose transported (1) • no/less CO_2 removed (1)		3
	(iii)	• takes blood beyond blockage (1) • to supply heart muscles (1)		2

Total for question = 8 marks

Type MS · Describe how heart disease is prevented and treated

Answer 5 of 7 · Question 3 part b.iii · 2 marks

2020 NOV · Paper 02 (regional R) · extract 2 of 2 · find part b.iii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Describe how heart disease is prevented and treated

Answer 6 of 7 · Question 9 part b · 4 marks

2022 JUN · Paper 01 (regional R) · find part b in this mark scheme extract

9 (a)	• two groups of people; • provide one group with statins and the other without; • measure/compare levels of cholesterol / results after a period of time;		3
(b)	Any four from the following: • catheter inserted (into affected area); • balloon/stent inflated/expanded; • stent expands; • blood vessel opened/diameter of vessel increased; • blood flows more freely; • more oxygen to muscle (tissue); • more aerobic respiration;		4

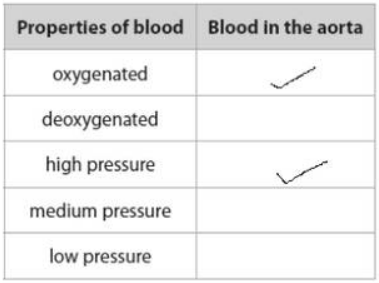
Total marks 7

Question number	Answer	Notes	Marks

Type MS · Describe how heart disease is prevented and treated

Answer 7 of 7 · Question 2 part b · 2 marks

2023 JAN · Paper 02 · extract 1 of 2 · find part b in this mark scheme extract

Number					
2	(a)	(i)	left ventricle;		1
		(ii)	Two from: • valve; • prevents the backflow of blood; • so the blood flows the correct way around the heart/in one direction;	Ignore names of valves	2
		(iii)	 Properties of blood / Blood in the aorta oxygenated ✓ deoxygenated high pressure ✓ medium pressure low pressure	One mark removed for every additional incorrect tick.	2
		(iv)	the lungs/pulmonary vein;		1
	(b)		Two from: • it could be diseased/damaged/fatty deposits/blocked arteries; • the tissue/blood type doesn't match; • it has been removed from person for too long; • wrong size;	accept named diseases allow risk of rejection	2
	(c)	(i)	C platelets;		1
		(ii)	• fibrinogen is not converted to fibrin/fibrin not produced; • so factor XIII has no substrate/fibrin cannot form the clot/mesh not formed;		2

Total 11 marks

Type MS · Describe how heart disease is prevented and treated

Answer 7 of 7 · Question 2 part b · 2 marks

2023 JAN · Paper 02 · extract 2 of 2 · find part b in this mark scheme extract

Question number	Answer	Notes	Marks